

Investigation of series CLR circuit driven into resonance

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Abstract—(Word Count: 1491) Resonance in driven electrical oscillators is a key concept in sensing and communications. This experiment tests the standard series CLR (capacitor-inductor-resistor) model by measuring the steady-state voltage across the capacitor as a function of driving frequency for three different series resistances. The resonance curves were obtained with an oscilloscope with built-in function generator, and fitted with the theoretical transfer function to find the resonance frequency, bandwidth and quality factor. The measured resonance frequencies, $f_{\text{peak}} \approx 22.8$ kHz, agree with values calculated from the nominal inductance, L and capacitance, C to within the quoted uncertainties. The measured bandwidths are larger and the quality factors smaller than predicted, and the fits yield reduced chi-squared goodness-of-fit statistic $\chi^2 \gg 1$, indicating additional damping not included in the ideal series model.

Index Terms—resonance, damping, CLR circuit

I. INTRODUCTION

Resonance is a universal feature of driven and damped oscillators. In an electrical setting, one example is the series CLR (capacitor-inductor-resistor) circuit.

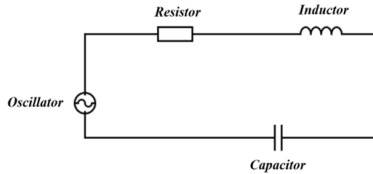


Fig. 1. Illustration of a generic CLR circuit

The charge in the CLR circuit as a function of time $q(t)$ obeys, from Kirchhoff's law combining with the relations together with $V_R = RI(t)$, $V_L = L \frac{dI(t)}{dt}$, $V_C = \frac{q(t)}{C}$ where V_R , V_L and V_C are the voltages across the resistor, inductor and capacitor, respectively, $q(t)$ is the charge on the capacitor, and $I(t) = \dot{q}(t)$ is the loop current.

This driven CLR equation, along with its mechanical-oscillator analogy, is widely documented in undergraduate physics literature [1], [2].

$$L \ddot{q} + R \dot{q} + \frac{q}{C} = E_0 \cos(\omega t) \quad (1)$$

where R is the series resistance, L is the inductance, C is the capacitance, $I(t) = \dot{q}(t)$ is the loop current, and the source

is $E(t) = E_0 \cos(\omega t)$ with peak amplitude E_0 (V) and angular frequency ω .

Taking the above equation as the starting point for modelling the steady-state response, we have a sinusoidal particular solution [3]:

$$q(t) = q_0 \cos(\omega t + \phi) \quad (2)$$

in which q_0 is the amplitude and ϕ is the fixed phase shift, in relation to the drive.

Substituting (2) into (1) yields the phase condition of

$$\tan \phi = \frac{R}{\omega L - \frac{1}{\omega C}}, \quad (3)$$

In this experiment, the voltage across the capacitor (V_C) is chosen to be the description of the circuit's response. The steady-state peak amplitude of V_C , denoted V_{C0} , can be obtained from the same substitution as

$$V_{C0} = \frac{E_0}{\sqrt{(1 - \omega^2 LC)^2 + (\omega RC)^2}}, \quad (4)$$

which we will fit to the measured resonance curve.

The undamped natural angular frequency ω_0 of the free oscillating circuit is:

$$\omega_0 = \frac{1}{\sqrt{LC}} \quad (5)$$

For a damped circuit, by differentiating Eq. (4) with respect to ω , it is noted that at an angular frequency given by:

$$\omega^2 = \omega_0^2 - \frac{R^2}{2L^2}, \quad (6)$$

the amplitude of V_C is maximised, where this condition is called *amplitude resonance*.

The sharpness of the amplitude resonance curve is quantified by the quality factor Q :

$$Q = \frac{\omega_{\text{peak}}}{\Delta\omega}, \quad (7)$$

where ω_{peak} is the angular frequency at which V_C reaches its maximum, and $\Delta\omega = \omega_2 - \omega_1$ is the bandwidth between the half-power points ω_1 and ω_2 , as shown in Fig. 2. Since $\omega = 2\pi f$, this relation can also be written approximately as

$$Q \approx \frac{f_{\text{peak}}}{\Delta f}, \quad (8)$$

where f_{peak} is the resonance frequency and $\Delta f = f_2 - f_1$ is the half-power bandwidth in Hz.

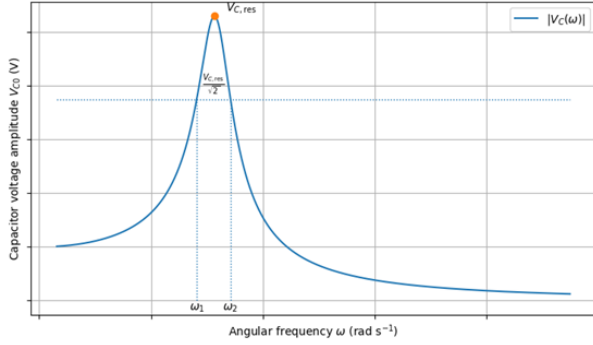


Fig. 2. Sketch of a typical voltage resonance curve, showing ω_1 and ω_2 , along with the corresponding half-power points

In the damped series case this is equivalent to:

$$Q = \frac{1}{R} \sqrt{\frac{L}{C}} = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 RC} \quad (9)$$

which links bandwidth directly to component values [2].

Physically, the resonance arises from a balance between energy stored alternately in the inductor's magnetic field and the capacitor's electric field, while the resistor continuously dissipates energy as heat. A high quality factor Q corresponds to weak damping, meaning energy circulates many times between L and C before being lost, producing a sharp and narrow resonance peak. Increasing the resistance strengthens dissipation, lowering Q and broadening the resonance because the system loses energy more quickly each cycle.

A wide array of technologies is associated with the same ω_0 , Q , bandwidth, and detuning concepts. Strongly coupled resonant coils only produce effective mid-range links in wireless power transfer when Q , $\Delta\omega$, and detuning $(\omega - \omega_0)$ are properly balanced [4]. Operating close to resonance in contactless electromagnetic sensing (such as magnetic-induction tomography) amplifies small inductive or conductive perturbations into detectable amplitude/phase shifts [5].

The aim of this experiment was to provide a quantitatively robust, uncertainty-aware validation of the driven-oscillator model relevant to power, sensing, and communication hardware. By measuring the capacitor voltage amplitude $V_C(\omega)$ and the phase response $\phi(\omega)$ across frequency for several resistor values R , the experiment aimed to characterise the resonance behaviour of the *CLR* circuit and extract ω_{peak} , the half-power bandwidth $\Delta\omega$, and the quality factor Q using the half-power method.

II. MATERIALS AND METHODS

This experiment used a signal source, measurement instruments, and passive components configured to construct a driven series *CLR* circuit while obtaining the capacitor voltage $V_C(t)$ as a function of frequency. A *KEYSIGHT DSOX1202G* two-channel oscilloscope captured the source voltage on CH (channel) 1 and V_C on CH 2, as shown in

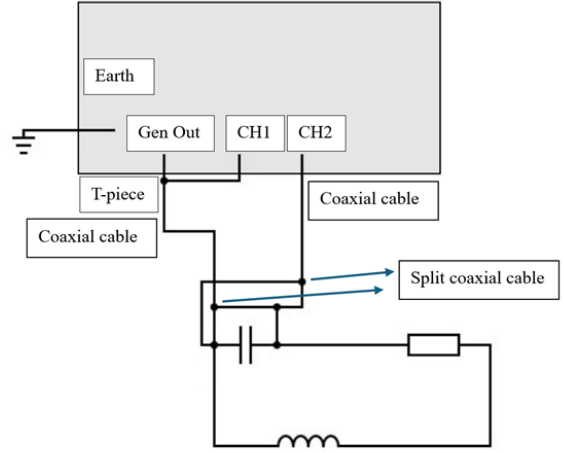


Fig. 3. Illustration of experiment set up, showing connections between the oscilloscope with built-in signal generator and the breadboard with *CLR* circuit

Fig. 3. Its built-in signal generator provided the sinusoidal drive.

The circuit was built on a breadboard, consisting of an inductor L with inductance of $22 \pm 1 \times 10^3 \mu\text{H}$, and a capacitor C with capacitance of $22 \pm 1 \times 10^2 \text{ pF}$. These were connected, one at a time, to resistors $R_1 = 1500 \pm 80 \Omega$, $R_2 = 500 \pm 30 \Omega$, and $R_3 = 150 \pm 8 \Omega$. The uncertainty of the components was calculated from the nominal tolerance of 5%.

Before data collection, the oscilloscope self-calibration was run. The generator offset was set to 0 V, the waveform set to sine, and the wave-generation amplitude set to $E_0 = 200 \text{ mV}$.

Amplitude–frequency measurements were then performed to obtain $V_C(f)$. The automated analysis function of the oscilloscope was used to aid accuracy, providing the possibility of obtaining up to 900 data points. The generator frequency f (Hz) was swept from $f_{\text{min}} = 5.500 \text{ kHz}$ to $f_{\text{max}} = 40.00 \text{ kHz}$, in accordance with the calculated theoretical resonance frequencies of $21.6 \pm 0.5 \text{ kHz}$, $22.7 \pm 0.5 \text{ kHz}$, and $22.9 \pm 0.5 \text{ kHz}$ for R_1 , R_2 , and R_3 , respectively, from Eq. (6).

The oscilloscope continuously recorded the gain of V_C at each frequency step, and the acquisition was exported directly as a comma-separated values (CSV) file containing frequency–amplitude pairs.

Those procedures were performed for each of the three resistors, and the exported data were then processed to determine the resonance characteristics.

Using the oscilloscope's analysis function, the voltage gain of the series *CLR* circuit, defined as

$$G(f) \equiv \frac{V_C(f)}{E_0}, \quad (10)$$

was evaluated.

For each sweep, the peak gain G_{peak} at f_{peak} was identified, then the half-power points $f_1 < f_{\text{peak}} < f_2$ were located where

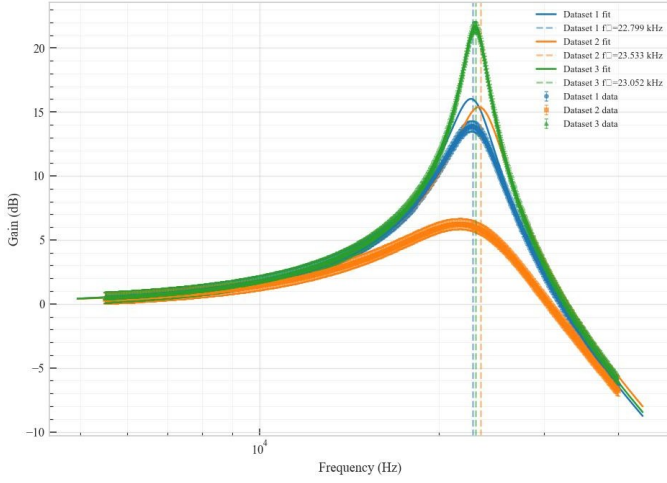


Fig. 4. Sketch of experimental and fitted voltage (gain) resonance curves of various levels of damping

$G = \frac{G_{\text{peak}}}{\sqrt{2}}$. Then the bandwidth $\Delta f = f_2 - f_1$ was obtained, and the quality factor Q was calculated from Eq. (9).

III. RESULTS AND ANALYSIS

The results are summarised in Tables 1 and 2, with uncertainties given to one significant figure, calculated by error propagation.

In the graphs, the sweeps using R_2 , R_1 and R_3 was referred as sweep A, sweep B and sweep C, respectively.

TABLE I
CALCULATED EXPERIMENTAL RESONANCE FREQUENCIES FOR EACH SWEEP

Resistor	f_{peak} (kHz)	f_1 (kHz)	f_2 (kHz)
R_1	22.446 ± 0.007	16.768 ± 0.008	25.681 ± 0.006
R_2	22.799 ± 0.003	19.570 ± 0.002	25.512 ± 0.001
R_3	23.001 ± 0.001	21.219 ± 0.006	24.791 ± 0.007

TABLE II
BANDWIDTH AND QUALITY FACTOR DERIVED FROM FREQUENCY MEASUREMENTS

Resistor	Δf (kHz)	Q
R_1	8.91 ± 0.01	2.438 ± 0.003
R_2	5.942 ± 0.002	3.821 ± 0.002
R_3	3.572 ± 0.009	6.44 ± 0.02

A. Effect of damping on electrical resonance

All three resonance frequencies fell in a tight band between 22–23 kHz, as shown in Fig. 4, Fig. 5 and Table I, proving the weak dependence of resonance frequency on resistance, since inductance and capacitance remain unchanged for the three sweeps.

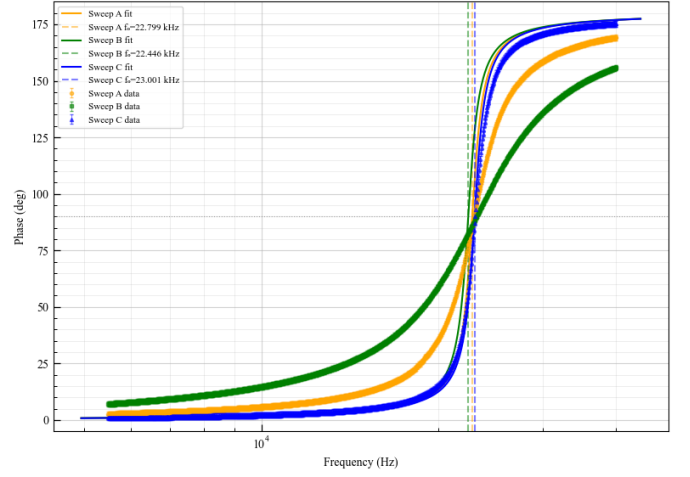


Fig. 5. Sketch of experimental phase resonance curves of various levels of damping

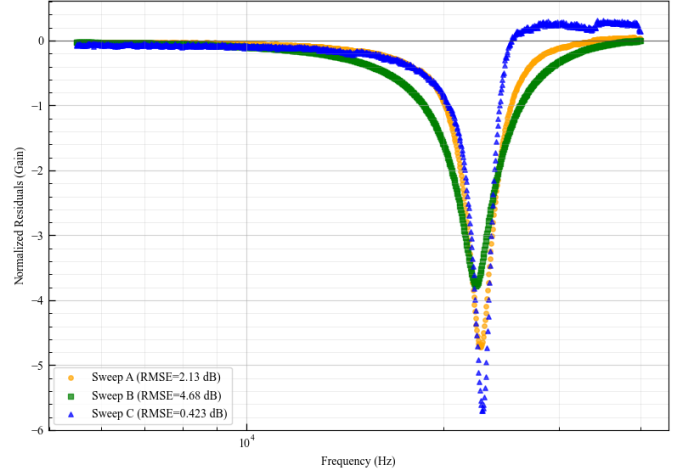


Fig. 6. Sketch of normalised gain residuals

The bandwidth Δf varied substantially between sweeps. Sweep using R_1 exhibits the lowest f_1 and the highest f_2 , and hence the largest bandwidth Δf , while the R_3 sweep has the narrowest Δf . Their bandwidths differ by more than a factor of two, further confirming the effect of the various resistances.

From Table I, between R_1 and R_3 sweeps, Q changes by a factor of approximately $6.44/2.438 \approx 2.64$. This increase in Q correlates directly with the change in bandwidth, confirming the relation $Q \approx f_{\text{peak}}/\Delta f$ derived from Eq. (7), showing that a higher quality factor corresponds to a narrower resonance.

B. Residual and χ^2 analysis

The residuals show that all three sweeps have the same qualitative behaviour, the residuals are close to zero at low frequencies, become the most negative near the resonance peak, and then rise back towards zero at high frequencies. This pattern shows that this model slightly overestimates the peak gain and slightly underestimates the others. The normalised

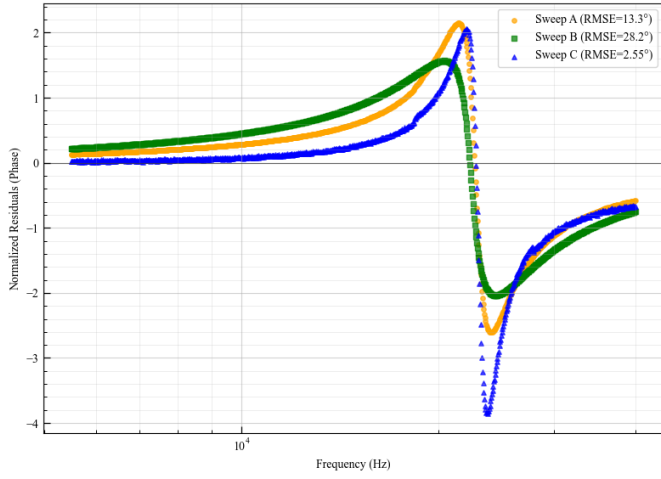


Fig. 7. Sketch of normalised phase residuals

residuals were obtained by dividing the residuals by their root-mean-square value (RMSE), yielding dimensionless quantities.

TABLE III
REDUCED χ^2 VALUES CALCULATED FROM EACH SWEEP, $\chi^2_{\nu, \text{GAIN}}$ AND $\chi^2_{\nu, \text{PHASE}}$ DENOTE THE REDUCED CHI-SQUARED FOR THE GAIN AND PHASE FITS, RESPECTIVELY.

Resistor	$\chi^2_{\nu, \text{gain}}$	$\chi^2_{\nu, \text{phase}}$
R_1	2200	800
R_2	450	180
R_3	18	6.5

A reduced χ^2 goodness-of-fit test was performed on each of the sweeps. The reduced chi-squared $\chi^2_{\nu} = \chi^2/\nu$ (χ^2 is the usual chi-squared statistic and ν is the number of degrees of freedom) was computed for each sweep.

The resulting reduced chi-squared values are much greater than 1 for all three sweeps. This indicates that the discrepancies between data and fit are dominated by systematic modelling errors, suggesting underfitting. Stray capacitances and lead inductances might be a reason of underfitting, which have not been considered in the calculations.

C. Agreement analysis with theoretical results

TABLE IV
THEORETICAL RESONANCE PARAMETERS CALCULATED FROM THE NOMINAL VALUES, SUBSCRIPTS 'TH' DENOTE THEORETICAL VALUES COMPUTED FROM THE NOMINAL COMPONENT VALUES.

Resistor	R / Ω	$f_{\text{peak, th}} / \text{kHz}$	$\Delta f_{\text{th}} / \text{kHz}$	Q_{th}
R_1	1500 ± 80	21.6 ± 0.5	10.8 ± 0.8	2.1 ± 0.2
R_2	500 ± 30	22.7 ± 0.5	3.6 ± 0.3	6.3 ± 0.5
R_3	150 ± 8	22.9 ± 0.5	1.1 ± 0.1	21 ± 2

Table IV presents the theoretical resonance parameters, calculated from Eqs. (5)–(8) using the nominal component values. The values are then compared with the experimental ones in Tables I and II.

For all three sweeps, the measured resonance frequencies f_{peak} agree well with the theoretical predictions. For example, the sweep using $R_2 = 500 \Omega$, yields $f_{\text{peak}} = 22.799 \pm 0.003 \text{ kHz}$, compared with $f_{\text{peak, th}} = 22.7 \pm 0.5 \text{ kHz}$; sweeps using R_1 and R_3 show similarly small deviations ($< 2\sigma$), where σ is the quoted standard uncertainty on the theoretical prediction.

In contrast, the agreement for the bandwidth Δf and the quality factor Q is poorer, especially for the lower-damping sweeps. Q_{th} increases from 2.1 ± 0.2 for $R_1 = 1500 \Omega$ to 6.3 ± 0.5 for $R_2 = 500 \Omega$ and 21 ± 2 for $R_3 = 150 \Omega$. Experimentally, however, the corresponding values are $Q = 2.438 \pm 0.003$, 3.821 ± 0.002 , and 6.44 ± 0.02 for R_1 , R_2 , and R_3 respectively. Thus, while the highest-resistance case (R_1) agrees with theory at the 2σ level, the lower-resistance sweeps show much smaller Q and larger Δf than predicted.

This systematic trend indicates an underestimation of the total damping in the circuit. The inductor's internal series resistance and additional loss channels in the wiring and breadboard could contribute extra resistance that does not appear in the nominal R values used to calculate Table IV. This interpretation is consistent with the large reduced χ^2 values and the systematic residual patterns discussed in Sec. III-B.

IV. CONCLUSION

The measurements successfully tested the driven series CLR resonance model. The experimental resonance frequencies for all three resistor values were consistent with those expected from the nominal L and C values within the quoted uncertainties. In contrast, the experimentally determined bandwidths Δf and quality factors Q were systematically broader and smaller than predicted, indicating that the simple model underestimates the total damping in the circuit and is therefore underfitting the data. This discrepancy can be explained by additional series resistance from the inductor and wiring, which is not included in the idealised circuit model. A more complete description would include these parasitic elements explicitly, for example by measuring the inductor impedance as a function of frequency, extracting an effective parasitic resistance, and refitting the data with an augmented circuit model. These conclusions are consistent with the broader literature on resonant electrical systems, which reports that real coils and wiring introduce additional dissipation beyond the idealised model [6].

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